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Cosmological models

In the previous chapter, we considered the geometric and kinematic properties of the Friedmann–Robertson–Walker (FRW) metric and derived the *cosmological field equations* for the scale factor $R(t)$. In this chapter, we will use the cosmological field equations to determine the behaviour of the scale factor as a function of cosmic time in various *cosmological models*.

15.1 Components of the cosmological fluid

In a general cosmological model, the universe is assumed to contain both matter and radiation. In addition, the cosmological constant Λ is generally assumed to be non-zero. As discussed in Section 8.7, the modern interpretation of Λ is in terms of the energy density of the vacuum, which may also be modelled as a perfect fluid (with a peculiar equation of state). Thus, one usually adopts the viewpoint that the cosmological fluid consists of *three* components, namely matter, radiation and the vacuum, each with a different equation of state. The total equivalent mass density is simply the sum of the individual contributions,

$$\rho(t) = \rho_m(t) + \rho_r(t) + \rho_\Lambda(t), \quad (15.1)$$

where t is the cosmic time and we have adopted the commonly used cosmological notation for the equivalent mass densities of matter, radiation and the vacuum respectively. Moreover, we shall assume that these three components are non-interacting (see Section 14.13); although matter and radiation did interact in the early universe, this is a reasonable approximation for most of its history.

As mentioned in Section 14.12, each component of the cosmological fluid is modelled as a perfect fluid with an equation of state of the form

$$p_i = w_i \rho_i c^2,$$

where the equation-of-state parameter w_i is a constant (and i labels the component). In particular $w_i = 0$ for pressureless ‘dust’, $w_i = \frac{1}{3}$ for radiation and $w_i = -1$ for the vacuum. In general, if w_i is a constant then, requiring that the weak energy condition is satisfied (see Exercise 8.8) and that the local sound speed $(dp/d\rho)^{1/2}$ is less than c , one finds that w_i must lie in the range $-1 \leq w_i \leq 1$. We see that this is indeed the case for dust, radiation and the vacuum. We now discuss each of these components in turn and conclude with a description of their relative contributions to the total density as the universe evolves.

Matter

In general, matter in the universe may come in several different forms. In addition to the normal baryonic matter of everyday experience (such as protons and neutrons), the universe may well contain more exotic forms of matter consisting of fundamental particles that lie beyond the ‘Standard Model’ of particle physics. Indeed, observations of the large-scale structure in the universe suggest that most of the matter is in the form of non-baryonic *dark matter*, which interacts electromagnetically only very weakly (and is hence invisible or ‘dark’). Moreover, dark matter may itself come in different forms, such as *cold dark matter* (CDM) and *hot dark matter* (HDM), the naming of which is connected to whether the typical energy of the particles is non-relativistic or relativistic. We shall not pursue this very interesting subject any further here¹ but merely note that the total matter density (at any particular cosmic time t) may be expressed as the sum of the baryonic and dark matter contributions,

$$\rho_m(t) = \rho_b(t) + \rho_{dm}(t).$$

In the following discussion, we will not differentiate between different types of matter, since it is only the total matter density that determines how the scale factor $R(t)$ evolves with cosmic time t . We shall also make the common assumption that the matter particles (in whatever form) have a thermal energy that is much less than their rest mass energy, and so the matter can be considered to be pressureless, i.e. *dust*. In this case the equation of state parameter is simply $w = 0$. Thus, from (14.44), if the matter has a present-day proper density of $\rho_m(t_0) \equiv \rho_{m,0}$, its density at some other cosmic time t is given by

$$\rho_m(t) = \rho_{m,0} \left[\frac{R_0}{R(t)} \right]^3 \quad \text{or} \quad \rho_m(z) = \rho_{m,0} (1+z)^3,$$

¹ For a full discussion, see (for example) T. Padmanabhan, *Structure Formation in the Universe*, Cambridge University Press, 1993.

where in the second expression we have used (14.17) to write the density in terms of the redshift z . These expressions concur with our expectation for the behaviour of the space density of dust particles in an expanding universe.

Radiation

The term radiation naturally includes photons, but also other species with very small or zero rest masses, such that they move relativistically today. An example of the latter is neutrinos (which may in fact have a small non-zero rest mass). The total equivalent mass density of radiation in the universe at some cosmic time t may then be written as the sum of the photon and neutrino contributions:

$$\rho_r(t) = \rho_\gamma(t) + \rho_\nu(t).$$

Once again, we will not differentiate between different types of radiation in our subsequent discussion, since it is only the total energy density that determines the behaviour of the scale factor. For radiation, in general, we have $w = \frac{1}{3}$. Thus, from (14.44), if the total radiation in the universe has a present-day energy density of $\rho_{r,0}c^2$ then, at other cosmic times,

$$\rho_r(t) = \rho_{r,0} \left[\frac{R_0}{R(t)} \right]^4 \quad \text{or} \quad \rho_r(z) = \rho_{r,0}(1+z)^4.$$

In this case, the variation in the space density of photons (for example) again goes as $(1+z)^3$, but there is an additional factor $1+z$ resulting from the cosmological redshift of each photon.

It is worth noting that, to a very good approximation, the dominant contribution to the radiation energy density of the universe is due to the photons of the cosmic microwave background (CMB). This radiation is (to a very high degree of accuracy) uniformly distributed throughout the universe and has a *blackbody* form. For blackbody radiation, the number density of photons with frequencies in the range $[\nu, \nu + d\nu]$ is given by

$$n(\nu, T) d\nu = \frac{8\pi\nu^2}{c^3 (e^{h\nu/kT} - 1)} d\nu, \quad (15.2)$$

where T is the ‘temperature’ of the radiation. Since the energy per unit frequency is simply $u(\nu, T) = n(\nu, T)h\nu$, the total equivalent mass density of the radiation is

$$\rho_r(T) = \frac{1}{c^2} \int_0^\infty u_\nu d\nu = \frac{aT^4}{c^2},$$

where $a = 4\pi^2 k_B^4 / (60\hbar^3 c^3)$ is the reduced Stefan–Boltzmann constant. Observations show that the CMB is characterised by a present-day temperature

$T_0 = 2.726 \text{ K}$, which corresponds to a total present-day number density $n_{\gamma,0} \approx 4 \times 10^8 \text{ m}^{-3}$. It is easily shown that the CMB photon energy distribution retains its general blackbody form as the universe expands. Thus, at any given cosmic time t , the temperature of the CMB radiation in the universe is given by

$$T(t) = T_0 \left[\frac{R_0}{R(t)} \right] \quad \text{or} \quad T(z) = T_0(1+z), \quad (15.3)$$

from which we see that the universe must have not only been denser in the past, but also ‘hotter’.

Vacuum

As mentioned above the vacuum can be modelled as a perfect fluid having an equation of state $p = -\rho c^2$, so that the fluid has a negative pressure. This corresponds to an equation of state parameter $w = -1$. Thus, from (14.44), we see that at any cosmic time t , we have

$$\rho_{\Lambda} = \rho_{\Lambda,0} = \frac{\Lambda c^2}{8\pi G}.$$

Thus, the energy density of the vacuum always has the same *constant* value.

Relative contributions of the components

On combining the above results, we find that the variation in the total equivalent mass density (15.1) may be written as

$$\rho(t) = \rho_{m,0} \left[\frac{R_0}{R(t)} \right]^3 + \rho_{r,0} \left[\frac{R_0}{R(t)} \right]^4 + \rho_{\Lambda,0}. \quad (15.4)$$

From this expression, we see that the relative contributions of matter, radiation and the vacuum to the total density vary as the universe evolves. The details clearly depend on the relative values of $\rho_{m,0}$, $\rho_{r,0}$ and $\rho_{\Lambda,0}$. Typically, however, one would expect radiation to dominate the total density when $R(t)$ is small. As the universe expands, the radiation energy density dies away the most quickly and matter becomes the dominant component. Finally, if the universe continues to expand then the matter density also dies away and the vacuum ultimately dominates the energy density. We conclude by noting that cosmologists often define the *normalised scale parameter*

$$a(t) \equiv \frac{R(t)}{R_0},$$

in terms of which the above results are more compactly written, since $a_0 = 1$ by definition. We shall make use of this parameter further in subsequent sections.

15.2 Cosmological parameters

In our very simplified model of the universe discussed above, its entire history is determined by only a handful of *cosmological parameters*. In particular, if one specifies the values of the equivalent mass densities $\rho_m(t_*)$, $\rho_r(t_*)$ and ρ_Λ at some particular cosmic time t_* then the value of each density, and hence the total density, is determined at all other cosmic times t . Thus, specifying these quantities is sufficient to determine the scale factor $R(t)$ at all cosmic times. It is most natural to take t_* to be the present-day cosmic time t_0 , and so the cosmological model is entirely fixed by specifying the three quantities

$$\rho_{m,0}, \quad \rho_{r,0}, \quad \rho_{\Lambda,0}.$$

It is, however, both convenient and common practice in cosmology to work instead in terms of alternative *dimensionless* quantities, usually called *density parameters* or simply *densities*, which are defined by

$$\Omega_i(t) \equiv \frac{8\pi G}{3H^2(t)} \rho_i(t) \quad (15.5)$$

where $H(t)$ is the Hubble parameter and the label i denotes ‘m’, ‘r’ or ‘ Λ ’. It is worth noting that $\Omega_\Lambda(t)$ is, in general, a function of cosmic time t (unlike ρ_Λ , which is a constant). In terms of these new dimensionless parameters, the cosmological model may thus be fixed by specifying the values of the four present-day quantities

$$H_0, \quad \Omega_{m,0}, \quad \Omega_{r,0}, \quad \Omega_{\Lambda,0}. \quad (15.6)$$

A major goal of observational cosmology is therefore to determine these quantities for our universe. Significant advances in the last decade mean that cosmologists now know these values to an accuracy of just a few per cent.² We simply note here that

$$H_0 \approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad \Omega_{m,0} \approx 0.3, \quad \Omega_{r,0} \approx 5 \times 10^{-5}, \quad \Omega_{\Lambda,0} \approx 0.7, \quad (15.7)$$

where the units of H_0 are those most commonly used in cosmology, in which $1 \text{ Mpc} \equiv 10^6 \text{ parsecs} \approx 3.09 \times 10^{22} \text{ m}$; in SI units, $H_0 \approx 2.27 \times 10^{-18} \text{ s}^{-1}$. Perhaps most astonishing is that the present-day energy density of the universe is dominated by the vacuum!

² How these observational advances have been achieved is discussed in, for example, J. Peacock, *Cosmological Physics*, Cambridge University Press, 1999 or P. Coles & F. Lucchin, *Cosmology: The Origin and Evolution of Cosmic Structure* (2nd edition), Wiley, 2002.

We also note, for completeness, that cosmologists define further analogous dimensionless density parameters for the individual contributions to the matter and the radiation. For example, Ω_b , Ω_{dm} and Ω_ν are commonly used to denote the dimensionless density of baryons, dark matter and neutrinos respectively. For our universe, cosmological observations suggest the present-day values

$$\Omega_{b,0} \approx 0.05, \quad \Omega_{dm,0} \approx 0.25, \quad \Omega_{\nu,0} \approx 0, \quad (15.8)$$

noting, in particular, that only around one-sixth of the matter density is in the form of the familiar baryonic matter. Moreover, the majority of the baryonic matter seems not to reside in ordinary (hydrogen-burning) stars; the contribution of such stars is only $\Omega_* \approx 0.008$. The values of the individual quantities (15.8) affect the astrophysical process occurring in the universe and have a profound influence on, for example, the formation of structure. For determining the overall expansion history of the universe, however, only the quantities (15.6) need be specified.

The reason for defining the densities (15.5) becomes clear when we rewrite the second of the cosmological field equations (14.36) in terms of them. Dividing this equation through by R^2 and noting that $H = \dot{R}/R$, we obtain

$$1 = \Omega_m + \Omega_r + \Omega_\Lambda - \frac{c^2 k}{H^2 R^2}, \quad (15.9)$$

where, for notational simplicity, we have dropped the explicit time dependence of the variables. Indeed, it is also common practice to define the *curvature density parameter*

$$\Omega_k(t) = -\frac{c^2 k}{H^2(t) R^2(t)}, \quad (15.10)$$

so that, at all cosmic times t , we have the elegant relation

$$\Omega_m + \Omega_r + \Omega_\Lambda + \Omega_k = 1. \quad (15.11)$$

It should be noted that, in cosmological models with positive spatial curvature ($k = 1$), the parameter Ω_k is negative. Moreover, if the cosmological constant Λ is negative then so too is the vacuum density parameter Ω_Λ . This behaviour should be contrasted with that of Ω_m and Ω_r , which are always positive.

From (15.9), we see that the values of Ω_m , Ω_r and Ω_Λ determine the spatial curvature of the universe in a simple fashion. We have three cases:

$$\begin{aligned} \Omega_m + \Omega_r + \Omega_\Lambda < 1 &\Leftrightarrow \text{negative spatial curvature } (k = -1) &\Leftrightarrow \text{'open'}, \\ \Omega_m + \Omega_r + \Omega_\Lambda = 1 &\Leftrightarrow \text{zero spatial curvature } (k = 0) &\Leftrightarrow \text{'flat'}, \\ \Omega_m + \Omega_r + \Omega_\Lambda > 1 &\Leftrightarrow \text{positive spatial curvature } (k = 1) &\Leftrightarrow \text{'closed'}. \end{aligned}$$

The above relations are valid at any cosmic time t but are most often applied to the present day, $t = t_0$. In particular, it is also clear from (15.9) that, although the density parameters Ω_m , Ω_r and Ω_Λ are all, in general, functions of cosmic time t , their sum cannot change sign. Thus, the universe cannot evolve from one form of the FRW geometry to another. We note that cosmologists often add to the plethora of density parameters by also defining the *total density parameter*

$$\Omega \equiv \Omega_m + \Omega_r + \Omega_\Lambda = 1 - \Omega_k, \quad (15.12)$$

which is related to the total equivalent mass density (15.1) by $\Omega = 8\pi G\rho/(3H^2)$. From (15.7), we see that for our universe $\Omega_0 \approx 1$ or equivalently $\Omega_{k,0} \approx 0$, and it is therefore close to being spatially flat ($k = 0$).

Finally, it is worth noting that, for any cosmological model to be spatially flat, one requires $\Omega = 1$ and it is common to describe the corresponding total equivalent mass density as the *critical density*, which is given by

$$\rho_{\text{crit}} \equiv \frac{3H^2}{8\pi G}.$$

Hence, for any given value of the Hubble parameter, this expression gives the total equivalent mass density required for the universe to be spatially flat. Since recent cosmological observations suggest that our universe is indeed close to spatially flat and, (15.7), that $H_0 \approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, one finds that the present-day total equivalent mass density in our universe is

$$\rho_{\text{crit},0} = \frac{3H_0^2}{8\pi G} \approx 9.2 \times 10^{-27} \text{ kg m}^{-3}.$$

As mentioned above, it is thought that only around 30 per cent of this equivalent mass density is in the form of matter and only around 5 per cent in the form of baryonic matter. Nevertheless, it is worth noting that $\rho_{\text{crit},0} \approx 5.5 \text{ protons m}^{-3}$, and so the critical density turns out to be extremely low by laboratory standards.³

15.3 The cosmological field equations

Since the cosmological model can be fixed by specifying the values of the quantities listed in (15.6), it is worthwhile rewriting the cosmological field equations (14.36) in terms of these parameters. Let us begin with the second field equation. Recalling that $H = \dot{R}/R$, this may be written

$$H^2 = \frac{8\pi G}{3} \left(\sum_i \rho_i \right) - \frac{c^2 k}{R^2},$$

³ The fact that this is a number of order unity is an accident of our choice of units!

where the label i includes matter, radiation and the vacuum. From (15.4), (15.5) and (15.10), we therefore find

$$H^2 = H_0^2 (\Omega_{m,0} a^{-3} + \Omega_{r,0} a^{-4} + \Omega_{\Lambda,0} + \Omega_{k,0} a^{-2}), \quad (15.13)$$

where we have written the result in terms of the dimensionless scale parameter $a = R/R_0$. It should be remembered that $\Omega_{k,0} = 1 - \Omega_{m,0} - \Omega_{r,0} - \Omega_{\Lambda,0}$ and may be considered merely as a convenient shorthand. It is also worth noting that, since $a = R/R_0 = (1+z)^{-1}$, equation (15.13) immediately yields an expression for the Hubble parameter $H(z)$ as a function of redshift z .

We now turn to the first cosmological field equation in (14.36). Multiplying this equation through by $R/\dot{R}^2$ and again noting that $H = \dot{R}/R$, we have

$$\frac{R\ddot{R}}{\dot{R}^2} = -\frac{4\pi G}{3H^2} \sum_i \rho_i (1 + 3w_i),$$

where the label i once more includes matter, radiation and the vacuum. The left-hand side is equal to minus the deceleration parameter q defined in (14.19). Thus, substituting the appropriate value of w_i for each component and using (15.5), one finds the neat relation

$$q = \frac{1}{2}(\Omega_m + 2\Omega_r - 2\Omega_\Lambda). \quad (15.14)$$

If desired, one can easily write this equation explicitly in terms of the present-day values of the density parameters by using the result (15.13) and the relation

$$\Omega_i = \Omega_{i,0} \left(\frac{H_0}{H} \right)^2 a^{-3(1+w_i)},$$

which holds generally for matter, radiation and the vacuum.

15.4 General dynamical behaviour of the universe

The cosmological field equations (15.13) and (15.14) allow us to determine the general dynamical behaviour and the spatial geometry of the universe for any given set of values for the parameters $\Omega_{m,0}$, $\Omega_{r,0}$ and $\Omega_{\Lambda,0}$. The observations (15.7) suggest that the present-day value of the radiation density $\Omega_{r,0}$ is significantly smaller than the matter and vacuum densities. It is therefore a reasonable approximation to neglect $\Omega_{r,0}$ and parameterise a universe like our own in terms of just $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$ (and H_0 , which is irrelevant for our discussion in this section).

Figure 15.1 presents a summary of the properties of FRW universes dominated by matter and vacuum energy (known as Lemaitre models) as a function of

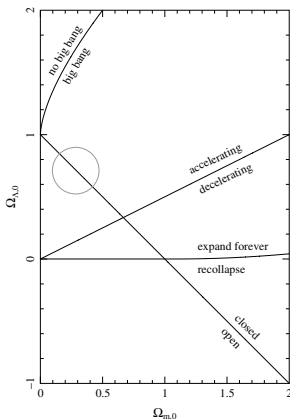


Figure 15.1 Properties of FRW universes dominated by matter and vacuum energy, as a function of the present-day density parameters $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$. The circle indicates the region of the parameter space that is consistent with recent cosmological observations.

position in the $(\Omega_{m,0}, \Omega_{\Lambda,0})$ parameter space. The dividing lines between the various regions may be determined from the field equations (15.13, 15.14) and the relation (15.11). In particular, the ‘open–closed’ line comes directly from (15.11) evaluated at the present epoch, which gives the condition

$$\Omega_{\Lambda,0} = 1 - \Omega_{m,0}.$$

Similarly, the ‘accelerating–decelerating’ line is obtained immediately by setting $q_0 = 0$ in (15.14) for $t = t_0$, which gives

$$\Omega_{\Lambda,0} = \frac{1}{2}\Omega_{m,0}.$$

The ‘expand–forever–recollapse’ line and the ‘big–bang–no–big–bang’ line require a little more work, as we now discuss.

In fact, both these lines are determined from the expression (15.13) for the Hubble parameter. In particular, the condition for the graph of $R(t)$, or equivalently

of $a(t)$, to have a turning point at some cosmic time $t = t_*$ is simply that $H(t_*) = 0$. Setting $\Omega_{k,0} = 1 - \Omega_{m,0} - \Omega_{\Lambda,0}$ in (15.13), we find that, after rearranging, this condition corresponds to

$$f(a) \equiv \Omega_{\Lambda,0} a^3 + (1 - \Omega_{m,0} - \Omega_{\Lambda,0})a + \Omega_{m,0} = 0. \quad (15.15)$$

This is a cubic equation for the value(s) of the scale factor $a = a_*$ at which $a(t)$ has a turning point. We are not, in fact, interested in the particular value(s) $a = a_*$ that solve (15.15) but only in whether a (real) solution exists in the region $a \geq 0$ (which is the only physically meaningful regime).

For the case $\Omega_{\Lambda,0} < 0$ we may deduce immediately from (15.14) that the universe must have started with a ‘big bang’, at which $a = 0$, and must eventually recollapse in a ‘big crunch’ as $a \rightarrow 0$ once more. In (15.14), a negative value of Ω_{Λ} means that the deceleration parameter q is always positive. Thus $\ddot{a}$ is always negative, and hence the $a(t)$ graph must be convex for all values of t . Since at the present epoch $\dot{a}(t_0) > 0$ (because we observe redshifts, not blueshifts), this means that $a(t)$ must have equalled zero at some point in the past, which it is usual to take as $t = 0$;⁴ similar reasoning may be used to deduce that the universe must eventually recollapse, although a little more care is required in this case. As the universe expands, the vacuum energy eventually dominates and so we need only consider the Ω_{Λ} -term on the right-hand side of (15.14), which will not tend to zero as the scale factor increases. Thus, $\ddot{a}$ cannot tend to zero and so $a \rightarrow 0$ at some finite cosmic time in the future.

In our further analysis, we now need only consider the case in which $\Omega_{\Lambda,0} \geq 0$ in (15.15), but this still requires some care. Let us first consider the case for which $\Omega_{\Lambda,0} = 0$. Immediately, we see that equation (15.15) then has the single solution $a_* = \Omega_{m,0}/(\Omega_{m,0} - 1)$, which is negative in the range $0 \leq \Omega_{m,0} \leq 1$, indicating that there is no (physically meaningful) turning point. Therefore, over this range, the ‘expand-forever–recollapse’ line is simply given by $\Omega_{\Lambda,0} = 0$. We must now address the far more complicated case for which $\Omega_{\Lambda,0} > 0$. In this case $f(a) \rightarrow \pm\infty$ as $a \rightarrow \pm\infty$. Moreover $f(0) = \Omega_{m,0}$, which is positive. Thus, for $f(a)$ to have a positive root, it must have a turning point in the region $a > 0$. On evaluating the derivatives $f'(a)$ and $f''(a)$ with respect to a , it is clear that, in the limiting case of interest, $f(a)$ must have the general form illustrated in Figure 15.2. Thus, we require $f(a_*) = f'(a_*) = 0$, which quickly yields

$$a_* = \left(\frac{\Omega_{m,0}}{2\Omega_{\Lambda,0}} \right)^{1/3}. \quad (15.16)$$

⁴ In fact, this reasoning is still valid in the case $\Omega_{\Lambda,0} = 0$, provided that the universe contains even an infinitesimal amount of matter (or radiation). Thus all cosmological models with $\Lambda \leq 0$ have a big-bang origin at some finite cosmic time in the past.

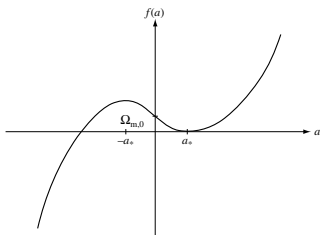


Figure 15.2 The limiting form of the cubic $f(a)$ defined in (15.15) for $\Omega_{\Lambda,0} > 0$.

On substituting this expression back into (15.15) one then obtains a separate cubic equation for $\Omega_{\Lambda,0}$, given by

$$4(1 - \Omega_{m,0} - \Omega_{\Lambda,0})^3 + 27\Omega_{m,0}^2\Omega_{\Lambda,0} = 0. \quad (15.17)$$

By introducing the variable $x = [\Omega_{\Lambda,0}/(4\Omega_{m,0})]^{1/3}$, this equation quickly reduces to

$$x^3 - \frac{3x}{4} + \frac{\Omega_{m,0} - 1}{4\Omega_{m,0}} = 0,$$

which is amenable to analysis using the standard formulae for finding the roots of a cubic. In particular, rewriting the resulting roots in terms of $\Omega_{\Lambda,0}$, one finds the following three cases:

- $0 < \Omega_{m,0} \leq \frac{1}{2}$, one positive root at

$$\Omega_{\Lambda,0} = 4\Omega_{m,0} \cosh^3 \left[\frac{1}{3} \cosh^{-1} \left(\frac{1 - \Omega_{m,0}}{\Omega_{m,0}} \right) \right]; \quad (15.18)$$

- $\frac{1}{2} < \Omega_{m,0} \leq 1$, one positive root at

$$\Omega_{\Lambda,0} = 4\Omega_{m,0} \cos^3 \left[\frac{1}{3} \cos^{-1} \left(\frac{1 - \Omega_{m,0}}{\Omega_{m,0}} \right) \right]; \quad (15.19)$$

- $\Omega_{m,0} > 1$, two positive roots, the larger given by (15.19) and the smaller by

$$\Omega_{\Lambda,0} = 4\Omega_{m,0} \cos^3 \left[\frac{1}{3} \cos^{-1} \left(\frac{1 - \Omega_{m,0}}{\Omega_{m,0}} \right) + \frac{4\pi}{3} \right]. \quad (15.20)$$

Moreover, from (15.16), one easily finds that $a_* < 1$ for (15.18, 15.19), whereas $a_* > 1$ for (15.20). Since the universe is expanding, $a_* < 1$ corresponds to a turning point in the past (i.e. no big bang), whereas $a_* > 1$ corresponds to a turning point in the future (i.e. recollapse).

The resulting lines, plotted in Figure 15.1, show some interesting features. In particular, we note that when $\Omega_{\Lambda,0} = 0$ there is a direct correspondence between the geometry of the universe and its eventual fate. In this case, open universes expand forever, whereas closed universes recollapse. This correspondence no longer holds in the presence of a non-zero cosmological constant, in which case any combination of spatial geometry and eventual fate is possible. It is also worth noting that the region of the $(\Omega_{m,0}, \Omega_{\Lambda,0})$ -plane consistent with recent cosmological observations is centred on the spatially flat model (0.3,0.7) and excludes the possibility of a zero cosmological constant at high significance. These observations also show the expansion of the universe to be accelerating. They also require the universe to have started at a big bang at some finite cosmic time in the past and to expand forever in the future.

15.5 Evolution of the scale factor

So far, we have considered only the limiting behaviour of the (normalised) scale factor $a(t)$ for different values of the cosmological parameters; this was summarised in Figure 15.1. We now discuss how to find the form of the $a(t)$ -curve at all cosmic times, for a given set of (present-day) cosmological parameter values. This behaviour is entirely determined by the cosmological field equation (15.13). Remembering that $H = \dot{a}/a$, this may be written as

$$\left(\frac{da}{dt}\right)^2 = H_0^2 (\Omega_{m,0} a^{-1} + \Omega_{r,0} a^{-2} + \Omega_{\Lambda,0} a^2 + 1 - \Omega_{m,0} - \Omega_{r,0} - \Omega_{\Lambda,0}). \quad (15.21)$$

Instead of working directly in terms of the cosmic time t , it is more convenient to introduce the new dimensionless variable

$$\hat{t} = H_0(t - t_0), \quad (15.22)$$

which measures cosmic time relative to the present epoch in units of the ‘Hubble time’ H_0^{-1} . In terms of this new variable, (15.21) becomes

$$\left(\frac{da}{d\hat{t}}\right)^2 = \Omega_{m,0} a^{-1} + \Omega_{r,0} a^{-2} + \Omega_{\Lambda,0} a^2 + 1 - \Omega_{m,0} - \Omega_{r,0} - \Omega_{\Lambda,0}. \quad (15.23)$$

There exist some special cases, where $\Omega_{m,0}$, $\Omega_{r,0}$ and $\Omega_{\Lambda,0}$ take on particular simple values, for which equation (15.23) can be solved analytically; we will

discuss some of these cosmological models in Section 15.6. In general, however, a numerical solution is necessary. Starting at the point $\hat{t} = 0$ (the present epoch), for which $a_0 = 1$, the normalised scale factor at time step $n + 1$ can be approximated by the Taylor expansion

$$a_{n+1} \approx a_n + \left(\frac{da}{d\hat{t}} \right)_n \Delta\hat{t} + \frac{1}{2} \left(\frac{d^2a}{d\hat{t}^2} \right)_n (\Delta\hat{t})^2, \quad (15.24)$$

where $\Delta\hat{t}$ is the (small) step size in $\hat{t}$. The coefficient of $\Delta\hat{t}$ is given by (15.23), and the coefficient of $(\Delta\hat{t})^2$ may be obtained by differentiating (15.23). The latter is important since, without the $(\Delta\hat{t})^2$ term, equation (15.24) would not carry the integration correctly through a value of a for which $da/d\hat{t}$ is small or zero.

Figure 15.3 shows the variation in the normalised scale factor $a(\hat{t})$ as a function of $\hat{t}$ for different values of $(\Omega_{m,0}, \Omega_{\Lambda,0})$ as indicated, assuming that $\Omega_{r,0}$ is negligible, as it is for our universe. In the top panel $\Omega_{m,0} + \Omega_{\Lambda,0} = 1$ in each case, so each universe has a flat spatial geometry ($k = 0$). The solid line corresponds to the case (0.3, 0.7), which is preferred by recent cosmological observations. An interesting cosmological ‘coincidence’ for this model is that the present epoch, $\hat{t} = 0$, corresponds almost exactly to the point of inflection on the $a(\hat{t})$ curve. A second such ‘coincidence’ is that the age of the universe in this model (i.e. the time since the big bang) is very close to one Hubble time.⁵ The broken-and-dotted line in the top panel of the figure corresponds to the case (0, 1), which is known as the *de Sitter model* and will be discussed further in Section 15.6. For the moment, we simply note that this model has no big-bang origin (although $a \rightarrow 0$ as $\hat{t} \rightarrow -\infty$) and will expand forever. The broken line in the top panel corresponds to the case (1, 0), which is known as the *Einstein–de Sitter model* and will also be discussed in Section 15.6. As we see from the figure, this model does have a big-bang origin. It is also on the borderline between expanding forever and recollapsing; it will in fact expand forever, but $\dot{a} \rightarrow 0$ as $\hat{t} \rightarrow \infty$.

In the bottom panel of Figure 15.3 we have $\Omega_{m,0} + \Omega_{\Lambda,0} \neq 1$ in each case, and so each universe is spatially curved; in particular the case (0.3, 0) is open and the cases (0.3, 2) and (4, 0) are closed. We see that the case (0.3, 2) has no big-bang origin, and is, in fact, what is known as a *bounce model*, where the universe collapses from large values of the scale factor and ‘bounces’ at some finite minimum value of a , after which it re-expands forever. Conversely, the case (4, 0) corresponds to a cosmological model with a big-bang origin that expands to some finite maximum value of a before recollapsing to a big crunch.

Before going on to discuss cosmological models that admit an analytic solution for $a(t)$, it is worth discussing the general case in the limit $a \rightarrow 0$. Whether

⁵ Whether such coincidences have some deeper significance is the subject of current cosmological research.

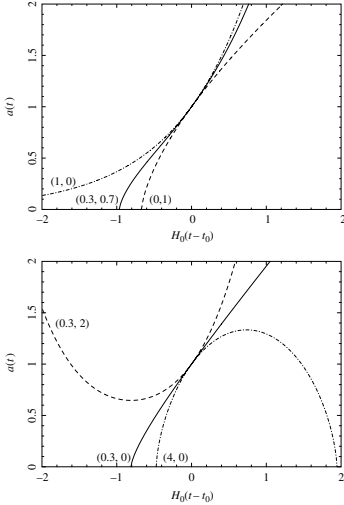


Figure 15.3 The variation in the normalised scale factor as a function of the dimensionless variable $H_0(t - t_0)$ for different values of $(\Omega_{m,0}, \Omega_{\Lambda,0})$ as indicated, assuming that $\Omega_{r,0}$ is negligible. Top panel: $\Omega_{m,0} + \Omega_{\Lambda,0} = 1$ in each case, so the universes have a flat spatial geometry ($k = 0$). Bottom panel: $\Omega_{m,0} + \Omega_{\Lambda,0} \neq 1$ in each case, so the universes are spatially curved; in particular, the case $(0.3, 0)$ is open and the cases $(0.3, 2)$ and $(4, 0)$ are closed.

considering the big bang or the big crunch, in this limit we can assume that the energy density of the universe is dominated by a one kind of source (which one will depend on the particular cosmological model under consideration). In this case, (15.23) can be written

$$\left(\frac{da}{dt}\right)^2 = \Omega_{i,0} a^{-(1+3w_i)} + \Omega_{k,0}, \quad (15.25)$$

where the label i denotes the dominant form of the energy density as $a \rightarrow 0$ and w_i is the corresponding equation-of-state parameter. Moreover, if we restrict

our attention to the (realistic) case, in which i denotes either dust ($w_i = 0$) or radiation ($w_i = \frac{1}{3}$), then the first term on the right-hand side of (15.25) dominates as $a \rightarrow 0$, so we can neglect the curvature density $\Omega_{k,0}$. In this case, (15.25) can be immediately integrated to give

$$a(\hat{t}) = \pm \left[\frac{3}{2}(1+w_i)\sqrt{\Omega_{i,0}} \right]^{2/[3(1+w_i)]} (\hat{t} - \hat{t}_*)^{2/[3(1+w_i)]}, \quad (15.26)$$

where $\hat{t}_*$ is the value of $\hat{t}$ at which $a = 0$ and the plus and minus signs correspond to the big bang and big crunch respectively. From (15.25), we also note that $da/d\hat{t} \rightarrow \infty$ as $a \rightarrow 0$. Thus, we conclude that the $a(\hat{t})$ -graph meets the $\hat{t}$ -axis at right angles.

15.6 Analytical cosmological models

Although in the general case the evolution of the (normalised) scale factor $a(t)$ must be determined numerically, there exist a number of special cases, corresponding to particular values of the cosmological parameters $\Omega_{m,0}$, $\Omega_{r,0}$ and $\Omega_{\Lambda,0}$, for which equation (15.23) can be solved analytically. We now discuss some of these analytical cosmological models, all of which have inherited special names that are widely used in cosmology. In this section, we will work in terms of the cosmic time t directly, rather than the dimensionless variable $\hat{t}$ defined in (15.22).

The Friedmann models

Cosmological models with a zero cosmological constant (and, strictly, a non-zero matter or radiation density) are known as the *Friedmann models*. As noted in Section 15.4, all Friedmann models have a big-bang origin at a finite cosmic time in the past. Moreover, it is possible to place a strict upper limit on the age of the universe in such models. Since the $a(t)$ -curve is everywhere convex, it is clear from Figure 15.4 that it crosses the t -axis at a time that is closer to the present time $t = t_0$ than the time at which the tangent to the point (t_0, a_0) reaches the t -axis (note that $a_0 = 1$). Clearly, the point where the tangent meets the t -axis is the point at which $a(t)$ would have been zero for $\dot{a} = \text{constant}$ and $\ddot{a} = 0$. The time elapsed from that point to the present epoch is simply $\dot{a}(t_0)/a(t_0) = H_0^{-1}$. Thus, in Friedmann models, the age of the universe must be less than the Hubble time:

$$t_0 < H_0^{-1}.$$

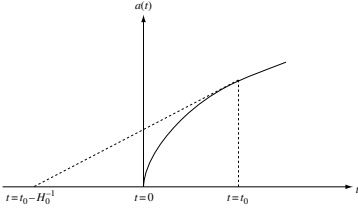


Figure 15.4 Diagram to illustrate that, for all Friedmann models, the age of the universe is less than the Hubble time $1/H_0$.

The behaviour of $a(t)$ near the big-bang origin is given by (15.26) and is independent of the curvature density parameter $\Omega_{k,0} = 1 - \Omega_{m,0} - \Omega_{r,0}$ (and hence of the sign of k). The future evolution, however, depends crucially on this constant. From (15.21) we can distinguish three possible histories, depending on the value of $\Omega_{k,0}$:

$$\begin{aligned} \Omega_{k,0} > 0 &\Leftrightarrow \text{open } (k = -1) &\Leftrightarrow \dot{a} \rightarrow \text{non-zero constant as } a \rightarrow \infty, \\ \Omega_{k,0} = 0 &\Leftrightarrow \text{flat } (k = 0) &\Leftrightarrow \dot{a} \rightarrow 0 \text{ as } a \rightarrow \infty, \\ \Omega_{k,0} < 0 &\Leftrightarrow \text{closed } (k = 1) &\Leftrightarrow \dot{a} = 0 \text{ at some finite value } a_{\max}. \end{aligned}$$

Thus, we see the main feature of Friedmann models, namely, that the *dynamics* of the universe is directly linked to its *geometry*. The three cases above are illustrated in Figure 15.5. We shall now find explicit analytical solutions for $a(t)$ in the special cases of a dust-only and a radiation-only Friedmann model. We will also obtain an analytic form for t as a function of a for the case of a spatially flat ($k = 0$) Friedmann model containing both matter and radiation.

Dust-only Friedmann models ($\Omega_{\Lambda,0} = 0, \Omega_{r,0} = 0$) In this case (15.21) becomes

$$\dot{a}^2 = H_0^2 (\Omega_{m,0} a^{-1} + 1 - \Omega_{m,0}) \quad \Rightarrow \quad t = \frac{1}{H_0} \int_0^a \left[\frac{x}{\Omega_{m,0} + (1 - \Omega_{m,0})x} \right]^{1/2} dx, \quad (15.27)$$

which may be integrated straightforwardly in each of the three cases $\Omega_{m,0} = 1$, $\Omega_{m,0} > 1$ and $\Omega_{m,0} < 1$ respectively, as follows.

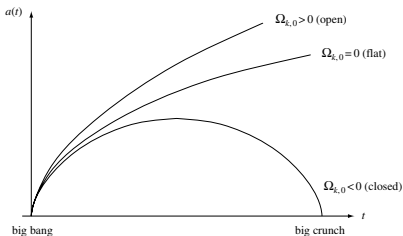


Figure 15.5 Schematic illustration of the evolution of the normalised scale factor $a(t)$ in closed, open and spatially flat Friedmann models.

- For $\Omega_{m,0} = 1$ ($k = 0$) the solution is immediate, and we find that

$$a(t) = \left(\frac{3}{2}H_0 t\right)^{2/3}. \quad (15.28)$$

This particular case is known as the *Einstein-de-Sitter* (or EdS) model.

- For $\Omega_{m,0} > 1$ ($k = 1$) the integral (15.27) can be evaluated by substituting $x = [\Omega_{m,0}/(\Omega_{m,0} - 1)] \sin^2(\psi/2)$, where ψ is known as the *development angle* and varies over the range $[0, \pi]$. One then obtains

$$a = \frac{\Omega_{m,0}}{2(\Omega_{m,0} - 1)}(1 - \cos \psi), \quad t = \frac{\Omega_{m,0}}{2H_0(\Omega_{m,0} - 1)^{3/2}}(\psi - \sin \psi),$$

which shows that the graph of $a(t)$ is a *cycloid*.

- For $\Omega_{m,0} < 1$ ($k = -1$) the integral (15.27) can be evaluated by substituting $x = [\Omega_{m,0}/(1 - \Omega_{m,0})] \sinh^2(\psi/2)$, and one obtains

$$a = \frac{\Omega_{m,0}}{2(1 - \Omega_{m,0})}(\cosh \psi - 1), \quad t = \frac{\Omega_{m,0}}{2H_0(1 - \Omega_{m,0})^{3/2}}(\sinh \psi - \psi).$$

In each case, one may also obtain expressions for $\rho_m(t) = \rho_{m,0}a^{-3}$ and $H(t) = \dot{a}/a$, and hence for $\Omega_m(t)$.

Radiation-only Friedmann models ($\Omega_{\Lambda,0} = 0, \Omega_{m,0} = 0$) In this case (15.21) becomes

$$\dot{a}^2 = H_0^2 (\Omega_{r,0} a^{-2} + 1 - \Omega_{r,0}) \quad \Rightarrow \quad t = \frac{1}{H_0} \int_0^a \frac{x}{\sqrt{\Omega_{r,0} + (1 - \Omega_{r,0})x^2}} dx, \quad (15.29)$$

which may again be integrated straightforwardly for $\Omega_{r,0} = 1$, $\Omega_{r,0} > 1$ and $\Omega_{r,0} < 1$ respectively.

- For $\Omega_{r,0} = 1$ ($k = 0$) the solution is again immediate, and we find that

$$a(t) = (2H_0 t)^{1/2}.$$

- For $\Omega_{r,0} < 1$ ($k = -1$) and $\Omega_{r,0} > 1$ ($k = 1$) the integral (15.29) can be evaluated by inspection to give

$$a(t) = \left(2H_0 \Omega_{r,0}^{1/2} t\right)^{1/2} \left(1 + \frac{1 - \Omega_{r,0}}{2\Omega_{r,0}^{1/2}} H_0 t\right)^{1/2}.$$

In each case, one may again obtain expressions for $\rho_r(t) = \rho_{m,0} a^{-3}$ and $H(t) = \dot{a}/a$, and hence for $\Omega_r(t)$.

Spatially flat Friedmann models ($\Omega_{\Lambda,0} = 0, \Omega_{m,0} + \Omega_{r,0} = 1$) In this case (15.21) becomes

$$\dot{a}^2 = H_0^2 (\Omega_{m,0} a^{-1} + \Omega_{r,0} a^{-2}) \quad \Rightarrow \quad t = \frac{1}{H_0} \int_0^a \frac{x}{\sqrt{\Omega_{m,0} x + \Omega_{r,0}}} dx, \quad (15.30)$$

which may be straightforwardly integrating by substituting $y = \Omega_{m,0} x + \Omega_{r,0}$ to obtain

$$H_0 t = \frac{2}{3\Omega_{m,0}^2} \left[(\Omega_{m,0} a + \Omega_{r,0})^{1/2} (\Omega_{m,0} a - 2\Omega_{r,0}) + 2\Omega_{r,0}^{3/2} \right].$$

Unfortunately, this expression cannot be easily inverted to give $a(t)$. Nevertheless, it is simple to show that the above expression becomes $\frac{2}{3}a^{3/2}$ for a matter-only model and $\frac{1}{2}a^2$ for a radiation-only model, and therefore agrees with our earlier results.

The Lemaitre models

The Lemaitre models are a generalisation of the Friedmann models in which the cosmological constant is non-zero. In particular, we will focus here on matter-only models ($\Omega_{r,0} = 0$), although our discussion is easily modified for radiation-only models, and can be extended to include models containing both matter and radiation. The general dynamical properties of Lemaitre models with $\Omega_{r,0} = 0$ were discussed in detail in Section 15.4, with particular focus on their limiting behaviour. We concentrate here on determining the generic form of the $a(t)$ -curve for models of this type that have a big-bang origin and will expand forever. We begin by considering the general case of arbitrary spatial curvature and then specialise to the spatially flat case. A model of the latter sort appears to provide a reasonable description of our own universe, if one neglects its radiation energy density.

Matter-only Lemaitre models with arbitrary spatial curvature ($\Omega_{r,0} = 0$) In this case the cosmological field equation (15.13) reads

$$\dot{a}^2 = H_0^2 (\Omega_{m,0} a^{-1} + \Omega_{\Lambda,0} a^2 + \Omega_{k,0}), \quad (15.31)$$

where $\Omega_{k,0} = 1 - \Omega_{m,0} - \Omega_{\Lambda,0}$. Obtaining explicit formulae giving, for example, the scale factor as a function of time is in general quite complicated, since the integrals turn out to involve elliptic functions,⁶ which are unfamiliar to most physicists these days. Nevertheless, we see that for small a the first term on the right-hand side dominates and the equation is easily integrated. Thus, after starting from a big-bang origin at $t = 0$, the $a(t)$ -curve at first increases as

$$a(t) = \left(\frac{3}{2} H_0 \sqrt{\Omega_{m,0}} t \right)^{2/3} \quad (\text{for small } t).$$

which agrees with our earlier result (15.26). As the universe expands, however, the matter energy density decreases and the vacuum energy eventually dominates. Thus, for large t (and hence large a), the second term on the right-hand side of (15.31) dominates. Once again the equation is then easily integrated to give

$$a(t) \propto \exp \left(H_0 \sqrt{\Omega_{\Lambda,0}} t \right) \quad (\text{for large } t).$$

⁶ See e.g. M. Abramowitz & I. A. Stegun, *Handbook of Mathematical Physics*, Dover, 1972.

From the above limiting behaviour at small and large t , it is clear that the universe must, at some point, make a transition from a decelerating to an accelerating phase. This occurs when $\ddot{a} = 0$, at which point the $a(t)$ -curve has a point of inflection. Differentiating (15.31), we find that

$$\ddot{a} = \frac{1}{2}H_0^2 (2\Omega_{\Lambda,0}a - \Omega_{m,0}a^{-2}). \quad (15.32)$$

From this result, we may verify immediately that at early cosmic times (when a is small) we have $\ddot{a} < 0$, and so the expansion is decelerating. As the universe expands, the deceleration gradually decreases until $\ddot{a}$ changes sign, after which the expansion accelerates ever more rapidly. We see that the value of the normalised scale factor at which the point of inflection ($\ddot{a} = 0$) occurs is given by

$$a_* = \left(\frac{\Omega_{m,0}}{2\Omega_{\Lambda,0}} \right)^{1/3}. \quad (15.33)$$

It is, in fact, possible to obtain an approximate analytic expression for the normalised scale factor $a(t)$ in the vicinity of the point of inflection. To do this, we must first obtain an approximate form for the cosmological field equation (15.31) in the vicinity of this point. Denoting the cosmic time at the point of inflection by t_* , we may perform separate Taylor expansions of a and $\dot{a}^2$ about $t = t_*$ to obtain

$$a \approx a_* + \dot{a}_*(t - t_*) \quad \text{and} \quad \dot{a}^2 \approx \dot{a}_*^2 + \ddot{a}_*\dot{a}_*(t - t_*)^2,$$

where, for notational convenience, we have written $a_* \equiv a(t_*)$, $\dot{a}_* \equiv \dot{a}(t_*)$, etc. Using the first expression to substitute for $(t - t_*)^2$ in the second, we obtain

$$\dot{a}^2 \approx \dot{a}_*^2 + \frac{\ddot{a}_*(a - a_*)^2}{\dot{a}_*}. \quad (15.34)$$

Differentiating (15.32) one easily obtains an expression for $\ddot{a}$. Then, substituting (15.33) into the resulting expression, and into (15.31), one finds that (15.34) becomes

$$\dot{a}^2 \approx H_0^2 [\Omega_{k,0} + 3\Omega_{\Lambda,0}a_*^2 + 3\Omega_{\Lambda,0}(a - a_*)^2].$$

This equation can now be integrated analytically and has the solution

$$a(t) = a_* + a_* \left[1 + \frac{1}{3}\Omega_{k,0} \left(\frac{1}{4}\Omega_{\Lambda,0}\Omega_{m,0}^2 \right)^{-1/3} \right]^{1/2} \sinh [H_0(3\Omega_{\Lambda,0})^{1/2}(t - t_*)].$$

(15.35)

An interesting property of this type of model is that in the case of positive spatial curvature ($k = 1$), for which $\Omega_{k,0} < 0$, there is a ‘coasting period’ in the

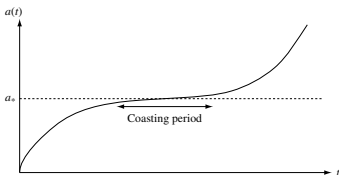


Figure 15.6 The behaviour of $a(t)$ in the Lemaitre model with $k = 1$. For $k = 0$ or $k = -1$, there is no extended coasting period.

vicinity of the point where $\ddot{a} = 0$, during which the value of $a(t)$ remains almost equal to a_* (see Figure 15.6). It is easily seen from (15.35) that, by setting the value of the quantity $\frac{1}{3}\Omega_{k,0}(\frac{1}{4}\Omega_{\Lambda,0}\Omega_{m,0}^2)^{-1/3}$ sufficiently close to -1 , one can make the coasting period arbitrarily long. Indeed, in the limiting case, it is easy to show that one requires that $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$ should satisfy (15.17).

Spatially flat matter-only Lemaitre models ($\Omega_{r,0} = 0$, $\Omega_{m,0} + \Omega_{\Lambda,0} = 1$) In this case one can give an explicit formula for the scale factor. Moreover, even if the universe turns out not to be exactly spatially flat, recent cosmological observations show that it is close enough to flatness for the formulae involved to act as a reasonable first approximation and so it is worthwhile to have them available.

In the spatially flat case, the cosmological field equation (15.13) may be written

$$\dot{a}^2 = H_0^2 [(1 - \Omega_{\Lambda,0})a^{-1} + \Omega_{\Lambda,0}a^2] \Rightarrow t = \frac{1}{H_0} \int_0^a \frac{x}{\sqrt{(1 - \Omega_{\Lambda,0}) + \Omega_{\Lambda,0}x^4}} dx.$$

This integral is a little more difficult than those considered earlier, but it can be made tractable by the substitution $y^2 = x^3|\Omega_{\Lambda,0}|/(1 - \Omega_{\Lambda,0})$, which yields

$$H_0 t = \frac{2}{3\sqrt{|\Omega_{\Lambda,0}|}} \int_0^{\sqrt{a^3|\Omega_{\Lambda,0}|/(1 - \Omega_{\Lambda,0})}} \frac{dy}{\sqrt{1 \pm y^2}},$$

where the plus sign in the integrand corresponds to the case $\Omega_{\Lambda,0} > 0$ and the minus sign to $\Omega_{\Lambda,0} < 0$. This may now be integrated easily to give

$$H_0 t = \frac{2}{3\sqrt{|\Omega_{\Lambda,0}|}} \begin{cases} \sinh^{-1} \left[\sqrt{a^3|\Omega_{\Lambda,0}|/(1 - \Omega_{\Lambda,0})} \right] & \text{if } \Omega_{\Lambda,0} > 0, \\ \sin^{-1} \left[\sqrt{a^3|\Omega_{\Lambda,0}|/(1 - \Omega_{\Lambda,0})} \right] & \text{if } \Omega_{\Lambda,0} < 0, \end{cases} \quad (15.36)$$

which may be inverted to give $a(t)$ in each case. One can also obtain analytic expressions for $H(t)$ and $\rho_m(t)$ (see Exercise 15.24) and thus for $\Omega_m(t)$ and $\Omega_\Lambda(t)$.

The de Sitter model

The de Sitter model is a particular special case of a Lemaitre model defined by the cosmological parameters $\Omega_{m,0} = 0$, $\Omega_{r,0} = 0$ and $\Omega_{\Lambda,0} = 1$. This model is therefore spatially flat ($k = 0$) but is not a true cosmological model in the strictest sense, since it assumes that the matter and radiation densities are zero. Nevertheless, it is interesting in its own right both for historical reasons and because of its close connection with the theory of inflation (see Section 16.1).

For the de Sitter model, the cosmological field equation (15.13) reads

$$\left(\frac{\dot{a}}{a}\right)^2 = H_0^2,$$

which immediately tells us that the Hubble parameter $H(t)$ is a constant and the normalised scale factor increases exponentially as

$$a(t) = \exp[H_0(t - t_0)] = \exp\left[\sqrt{\Lambda/3}c(t - t_0)\right],$$

where, in the second equality, we have expressed the solution in terms of the cosmological constant Λ . Thus, the de Sitter model has no big-bang singularity at a finite time in the past.

Einstein's static universe

All the cosmological models that we have constructed so far are *evolving* cosmologies. We know now, of course, that the universe is expanding and so there is no conflict with the field equations. Nevertheless, it is interesting historically to look at Einstein's static model of the universe. Einstein derived his field equations well before the discovery of the expansion of the universe and he was worried that he could not find static cosmological solutions. He therefore introduced the cosmological constant with the sole purpose of constructing static solutions.

For $\Lambda > 0$, we seek a solution to the field equations in which the universe is static, i.e. $\ddot{a} = \dot{a} = 0$. In this case, the Hubble parameter H is zero always, and so the dimensional densities in (15.5) are formally infinite. It is more convenient therefore to work with the field equations in their original forms (14.36). We see immediately that we require

$$4\pi G\rho_{m,0} = \Lambda c^2 = \frac{c^2 k}{R_0^2}.$$

In fact, the first equality can be more succinctly written as $\rho_{m,0} = 2\rho_{\Lambda,0}$. Since Λ is positive we thus require $k = 1$, and so the universe has positive spatial curvature.

How well did Einstein's static universe fit with cosmological observations of the time? The mean matter density of the universe is still a matter of great debate, but recent cosmological observations suggest that

$$\rho_{m,0} \approx 3 \times 10^{-27} \text{ kg m}^{-3}.$$

In Einstein's time, this value was estimated only to within about two orders of magnitude. Nevertheless, adopting the above value of $\rho_{m,0}$ we find that the scale factor is $R_0 \approx 2 \times 10^{26} \text{ m} \approx 6000 \text{ Mpc}$, which is more than sufficient for the closed spatial geometry to be large enough to encompass the observable universe. Also $\Lambda = 1/R_0^2 = 2.5 \times 10^{-53} \text{ m}^{-2}$, which is small enough to evade the limits on Λ from Solar System experiments ($|\Lambda| \leq 10^{-46} \text{ m}^{-2}$). Thus the Einstein static universe was not immediately and obviously wrong.

However, aside from the fact that the model disagreed with later observations indicating an expanding universe, it has the theoretically undesirable feature of being *unstable*. The cosmological constant must be fine-tuned to match the density of the universe. Thus, if we add or subtract one proton from this universe, or convert some matter into radiation, we will disturb the finely tuned balance between gravity and the cosmological constant and the universe will begin to expand or contract.

15.7 Look-back time and the age of the universe

Since the cosmological model may be fixed by specifying the values of the four (present-day) cosmological parameters H_0 , $\Omega_{m,0}$, $\Omega_{r,0}$ and $\Omega_{\Lambda,0}$, it is possible to use these quantities to determine other useful *derived* cosmological parameters. In this section we consider the look-back time and the age of the universe.

In Chapter 14, we showed that if a comoving particle (galaxy) emitted a photon at cosmic time t that is received by an observer at $t = t_0$ then the 'look-back time' $t_0 - t$ is given as a function of the photon's redshift by

$$t_0 - t = \int_0^z \frac{d\bar{z}}{(1 + \bar{z})H(\bar{z})}. \quad (15.37)$$

From the cosmological field equation (15.13), on noting that $a = R/R_0 = (1 + z)^{-1}$ we obtain the useful result

$$H^2(z) = H_0^2 [\Omega_{m,0}(1 + z)^3 + \Omega_{r,0}(1 + z)^4 + \Omega_{\Lambda,0} + \Omega_{k,0}(1 + z)^2]. \quad (15.38)$$

Thus, the look-back time to a comoving object with redshift z is given by

$$t_0 - t = \frac{1}{H_0} \int_0^z \frac{d\bar{z}}{(1+\bar{z}) \sqrt{\Omega_{m,0}(1+\bar{z})^3 + \Omega_{r,0}(1+\bar{z})^4 + \Omega_{\Lambda,0} + \Omega_{k,0}(1+\bar{z})^2}}.$$

We note that the differential form of this relation is perhaps more useful since one is often interested simply in the cosmic time interval dt corresponding to an interval dz in redshift. In any case, a more convenient form of the integral for evaluation is obtained by making the substitution $x = (z+1)^{-1}$, which yields

$$t_0 - t = \frac{1}{H_0} \int_{(1+z)^{-1}}^1 \frac{x dx}{\sqrt{\Omega_{m,0}x + \Omega_{r,0} + \Omega_{\Lambda,0}x^4 + \Omega_{k,0}x^2}}. \quad (15.39)$$

Assuming $\Omega_{r,0} = 0$ (which is a reasonable approximation for our universe), in Figure 15.7 we plot $H_0(t - t_0)$, the look-back time in units of the Hubble time, as a function of redshift for several values of $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$.

In any cosmological model with a big-bang origin, an extremely important quantity is the *age of the universe*, i.e. the cosmic time interval between the point when $a(t) = 0$ and the present epoch $t = t_0$. Since $z \rightarrow \infty$ at the big bang, we may immediately obtain an expression for the age of the universe in such a model

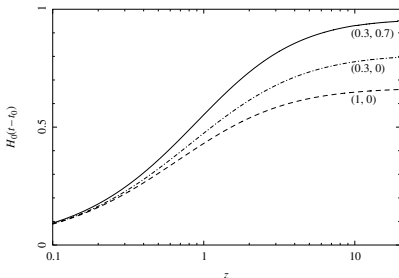


Figure 15.7 The variation in look-back time, in units of the Hubble time, as a function of redshift z for several sets of values $(\Omega_{m,0}, \Omega_{\Lambda,0})$ as indicated, assuming that $\Omega_{r,0}$ is negligible.

Table 15.1 *The age of the universe in Gyr for various cosmological models (with $\Omega_{r,0} = 0$)*

$\Omega_{m,0}$	$\Omega_{\Lambda,0}$	H_0 in $\text{km s}^{-1} \text{Mpc}^{-1}$		
		50	70	90
1.0	0.0	13.1	9.3	7.2
0.3	0.0	15.8	11.3	8.8
0.3	0.7	18.9	13.5	10.5

by letting $z \rightarrow \infty$ in (15.39), so that the lower limit of the integral equals zero. Since the resulting integral is dimensionless, we can write

$$t_0 = \frac{1}{H_0} f(\Omega_{m,0}, \Omega_{r,0}, \Omega_{\Lambda,0}),$$

where f is the value of the integral, which is typically a number of order unity. The age of the universe is therefore the Hubble time multiplied by a number of order unity. For general values of the density parameters $\Omega_{m,0}$, $\Omega_{r,0}$ and $\Omega_{\Lambda,0}$, it is not possible to perform the integral analytically and so one has to resort to numerical integration. Table 15.1 lists the age of the universe t_0 for the same values of $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$ as considered in Figure 15.7. It is interesting to compare these values with estimates of the ages of the oldest stars in globular clusters,

$$t_{\text{stars}} \approx 11.5 \pm 1.3 \text{ Gyr},$$

where the uncertainty is dominated by uncertainties in the theory of stellar evolution. Clearly, one requires $t_0 > t_{\text{stars}}$ for a viable cosmology!

It is worth noting that, in our discussion of analytical cosmological models in the previous section, we have already performed (a generalised version of) the relevant integral required to calculate the corresponding age of the universe in each case. Thus, for each model with a big-bang origin for which we have calculated an analytical form for $a(t)$ or $t(a)$, the corresponding age of the universe is obtained simply by setting $t = t_0$ and $a = 1$. For example, from (15.28), the age of an Einstein–de-Sitter universe is simply $t_0 = 2/(3H_0)$. Similarly, from (15.36), the age of a spatially flat matter-only Lemaitre model with $\Omega_{\Lambda,0} > 0$ is given by

$$t_0 = \frac{2}{3H_0\sqrt{\Omega_{\Lambda,0}}} \sinh^{-1} \sqrt{\frac{\Omega_{\Lambda,0}}{1-\Omega_{\Lambda,0}}} = \frac{2}{3H_0} \frac{\tanh^{-1} \sqrt{\Omega_{\Lambda,0}}}{\sqrt{\Omega_{\Lambda,0}}},$$

where, in the second equality, we have rewritten the result in a more useful form involving $\Omega_{\Lambda,0}$, using standard formulae for inverse hyperbolic trigonometric functions.

15.8 The distance–redshift relation

We may also obtain a general expression for the comoving χ -coordinate of a galaxy emitting a photon at time t that is received at time t_0 with redshift z . This is given by

$$\chi = \int_t^{t_0} \frac{c d\bar{t}}{R(\bar{t})} = \frac{c}{R_0} \int_0^z \frac{d\bar{z}}{H(\bar{z})}.$$

We may now substitute for $H(z)$ using the expression (15.38) derived in the previous section. Thus the χ -coordinate of a comoving object with redshift z is given by

$$\chi(z) = \frac{c}{R_0 H_0} \int_0^z \frac{d\bar{z}}{\sqrt{\Omega_{m,0}(1+\bar{z})^3 + \Omega_{r,0}(1+\bar{z})^4 + \Omega_{\Lambda,0} + \Omega_{k,0}(1+\bar{z})^2}}. \quad (15.40)$$

Once again, the differential form of this result is perhaps more useful, since one is often interested in the comoving coordinate interval $d\chi$ corresponding to an interval dz in redshift. As before, a simpler form for the integral is obtained by making the substitution $x = (1+\bar{z})^{-1}$, which yields

$$\chi(z) = \frac{c}{R_0 H_0} \int_{(1+z)^{-1}}^1 \frac{dx}{\sqrt{\Omega_{m,0}x + \Omega_{r,0} + \Omega_{\Lambda,0}x^4 + \Omega_{k,0}x^2}}. \quad (15.41)$$

From (14.29) and (14.31), the corresponding luminosity distance $d_L(z)$ and angular diameter distance $d_A(z)$ to the object are given by

$$d_L(z) = R_0(1+z)S(\chi(z)) \quad \text{and} \quad d_A(z) = \frac{R_0}{1+z}S(\chi(z)),$$

where $S(\chi)$ is given by (14.12), whereas the proper distance to the object is simply $d(z) = R_0 S(\chi(z))$. It is useful to introduce the notation $\chi(z) = cE(z)/(R_0 H_0)$, so that $E(z)$ denotes the integral in (15.41). Using the expression (15.10) to obtain $\Omega_{k,0}$, one can then write

$$R_0 S(\chi(z)) = \frac{c}{H_0} \begin{cases} |\Omega_{k,0}|^{-1/2} S(\sqrt{|\Omega_{k,0}|} E(z)) & \text{for } \Omega_{k,0} \neq 0, \\ E(z) & \text{for } \Omega_{k,0} = 0, \end{cases}$$

which allows simple direct evaluation of $d_L(z)$ and $d_A(z)$ in each case.

As was the case in the previous section, for general values of $\Omega_{m,0}$, $\Omega_{r,0}$ and $\Omega_{\Lambda,0}$ it is not possible to perform the integral (15.41) analytically and so one has to resort to numerical integration. Figure 15.8 shows plots of dimensionless luminosity distance $(c/H_0)^{-1}d_L(z)$ (top panel) and dimensionless angular diameter distance $(c/H_0)^{-1}d_A(z)$ (bottom panel) for various values of $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$, assuming that $\Omega_{r,0}$ is negligible; the solid, broken and dotted lines correspond to

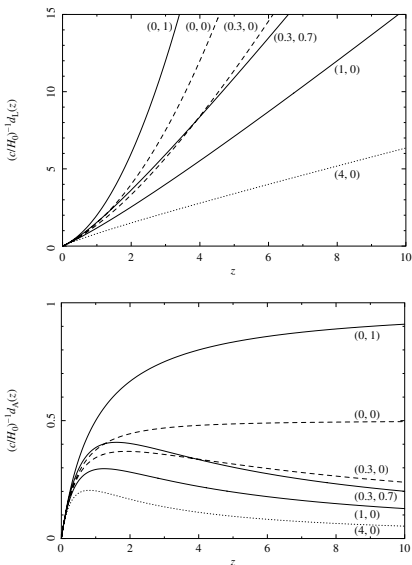


Figure 15.8 The variation in dimensionless luminosity distance (top panel) and dimensionless angular diameter distance (bottom panel) as functions of redshift, for different sets of values $(\Omega_{m,0}, \Omega_{\Lambda,0})$ as indicated, assuming that $\Omega_{r,0}$ is negligible. The solid, broken and dotted lines correspond to spatially flat, open and closed models respectively.

spatially flat, open and closed models respectively. In particular, it is worth noting that, for the models with a non-zero matter density, the angular diameter distance has a maximum at some finite value of the redshift $z = z_*$. Thus, for a source of fixed proper length ℓ , the angular diameter $\Delta\theta = \ell/d_A$ declines with redshift for $z < z_*$, as one might naively expect, but then *increases* with redshift for $z > z_*$. A very-high-redshift galaxy (if such a thing existed) would therefore cast a large,

but dim, ghostly image on the sky. The physical reason for this is that the light from a distant object was emitted when the universe was much younger than it is now – the object was close to us when the light was emitted. This, coupled with gravitational focussing of the light rays by the intervening matter in the universe, means that the galaxy looks big!

The integral (15.41) can, in fact, be evaluated analytically in some simple cases. As an example, consider the Einstein–de-Sitter (EdS) model ($\Omega_{m,0} = 1$, $\Omega_{r,0} = 0$, $\Omega_{\Lambda,0} = 0$). In this case, we find that

$$\chi(z) = \frac{c}{R_0 H_0} \int_{(1+z)^{-1}}^1 \frac{dx}{\sqrt{x}} = \frac{2c}{R_0 H_0} [1 - (1+z)^{-1/2}].$$

Thus, the luminosity distance in the EdS model is given as a function of z by

$$d_L(z) = \frac{2c}{H_0} (1+z) [1 - (1+z)^{-1/2}],$$

and the angular diameter distance by

$$d_A(z) = \frac{2c}{H_0} \frac{1}{1+z} [1 - (1+z)^{-1/2}].$$

Note that, in this case, $d_A(z)$ has a maximum at a redshift $z = 5/4$.

The relations between redshift and luminosity distance (angular diameter distance), form the basis of observational tests of the geometry of the universe. All one needs is a standard candle (for application of the luminosity-distance–redshift relation) or a standard ruler (for application of the angular-diameter-distance–redshift relation). Comparison with the predicted relations shown in Figure 15.8 can then fix the values of $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$. Unfortunately, standard candles and standard rulers are hard to find in the universe! Nevertheless, in recent years there has been remarkable progress, using distant Type Ia supernovae as standard candles and anisotropies in the cosmic microwave background radiation as a standard ruler. The results of these observations suggest that we live in a spatially flat universe with $\Omega_{m,0} \approx 0.3$ and $\Omega_{\Lambda,0} \approx 0.7$.

15.9 The volume–redshift relation

In Section 14.10 we found that, at the present cosmic time t_0 , the proper volume of the region of space lying in the infinitesimal coordinate range $\chi \rightarrow \chi + d\chi$ and subtending an infinitesimal solid angle $d\Omega = \sin \theta d\theta d\phi$ at the observer is

$$dV_0 = \frac{cR_0^2 S^2(\chi(z))}{H(z)} dz d\Omega, \quad (15.42)$$

the corresponding volume of this region at a redshift z being given by $dV(z) = dV_0/(1+z)^3$. We may now express dV_0 in terms of the cosmological parameters H_0 , $\Omega_{m,0}$, $\Omega_{r,0}$ and $\Omega_{\Lambda,0}$. Using the expressions (15.40), (15.38) and (15.10) for $\chi(z)$, $H(z)$ and Ω_k respectively, we find immediately that

$$dV_0 = \frac{(cH_0^{-1})^3}{h(z)} \begin{cases} |\Omega_{k,0}|^{-1} S^2(\sqrt{|\Omega_{k,0}|}E(z)) & \text{for } \Omega_{k,0} \neq 0, \\ E^2(z) & \text{for } \Omega_{k,0} = 0, \end{cases} \quad (15.43)$$

where we have defined the new function

$$h(z) \equiv \frac{H(z)}{H_0} = \sqrt{\Omega_{m,0}(1+z)^3 + \Omega_{r,0}(1+z)^4 + \Omega_{\Lambda,0} + \Omega_{k,0}(1+z)^2}$$

and $E(z) \equiv \int_0^z d\bar{z}/h(\bar{z})$ is the function defined in the previous section.

For general values of $\Omega_{m,0}$, $\Omega_{r,0}$ and $\Omega_{\Lambda,0}$, one must once again resort to numerical integration to obtain dV_0 . In Figure 15.9, we plot the dimensionless differential comoving volume element $(c/H_0)^{-3} dV_0/(dz d\Omega)$ as a function of redshift z for several values of $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$, assuming that $\Omega_{r,0} = 0$. In particular, we note that, in the currently favoured case $(\Omega_{m,0}, \Omega_{\Lambda,0}) = (0.3, 0.7)$, we may explore a large comoving volume by observing objects in the redshift range $z = 2-3$.

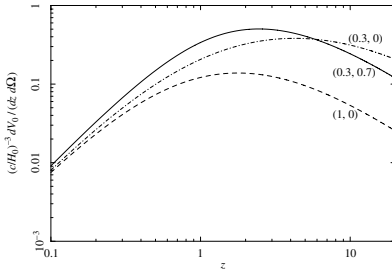


Figure 15.9 The variation in the dimensionless differential comoving volume element as a function of redshift z for several sets of values $(\Omega_{m,0}, \Omega_{\Lambda,0})$ as indicated, assuming that $\Omega_{r,0}$ is negligible.

15.10 Evolution of the density parameters

For the majority of our discussion so far, we have concentrated on exploring cosmological models with properties determined by fixing the values of the *present-day* densities $\Omega_{m,0}$, $\Omega_{r,0}$ and $\Omega_{\Lambda,0}$. From the definition (15.5), however, it is clear that each density is, in general, a function of cosmic time t . It is therefore of interest to investigate the evolution of these densities as the universe expands.

From (15.5) we have

$$\Omega_i(t) = \frac{8\pi G}{3H^2(t)} \rho_i(t) \quad \Rightarrow \quad \dot{\Omega}_i = \frac{8\pi G}{3H^2} \left(\dot{\rho}_i - \frac{2\dot{H}}{H} \rho_i \right), \quad (15.44)$$

where the label i denotes ‘m’, ‘r’ or ‘ Λ ’ and the dots denote differentiation with respect to cosmic time t . From the equation of motion (14.39) for a cosmological fluid, however, we have

$$\dot{\rho}_i = -3(1 + w_i)H\rho_i,$$

where we have written $H = \dot{R}/R$, and $w_i = p_i/(\rho_i c^2)$ is the equation-of-state parameter. Thus (15.44) becomes

$$\dot{\Omega}_i = -\Omega_i H \left[3(1 + w_i) + \frac{2\dot{H}}{H^2} \right], \quad (15.45)$$

where we have taken a factor of H outside the brackets for later convenience. We now need an expression for $\dot{H}$, which is given by

$$\dot{H} = \frac{d}{dt} \left(\frac{\dot{R}}{R} \right) = \frac{\ddot{R}}{R} - \left(\frac{\dot{R}}{R} \right)^2 = \frac{\ddot{R}}{R} - H^2,$$

and so we may write

$$\frac{\dot{H}}{H^2} = \frac{R\ddot{R}}{\dot{R}^2} - 1 = -(q + 1),$$

where q is the deceleration parameter. Substituting this result into (15.45) and using the expression (15.14) for q , we finally obtain the neat relation

$$\dot{\Omega}_i = \Omega_i H (\Omega_m + 2\Omega_r - 2\Omega_\Lambda - 1 - 3w_i).$$

Setting $w_i = 0, \frac{1}{3}$ and -1 respectively for matter (dust), radiation and the vacuum, we thus obtain

$$\begin{aligned} \dot{\Omega}_m &= \Omega_m H [(\Omega_m - 1) + 2\Omega_r - 2\Omega_\Lambda], \\ \dot{\Omega}_r &= \Omega_r H [\Omega_m + 2(\Omega_r - 1) - 2\Omega_\Lambda], \\ \dot{\Omega}_\Lambda &= \Omega_\Lambda H [\Omega_m + 2\Omega_r - 2(\Omega_\Lambda - 1)]. \end{aligned}$$

(15.46)

By dividing these equations by one another, we may remove the dependence on the Hubble parameter H and the cosmic time t and hence obtain a set of coupled first-order differential equations in the variables Ω_m , Ω_r and Ω_Λ alone. Therefore, given some general point in this parameter space, these equations define a unique trajectory that passes through this point. As an illustration, let us consider the case in which $\Omega_r = 0$. Dividing the remaining two equations then gives

$$\frac{d\Omega_\Lambda}{d\Omega_m} = \frac{\Omega_\Lambda[\Omega_m - 2(\Omega_\Lambda - 1)]}{\Omega_m[(\Omega_m - 1) - 2\Omega_\Lambda]},$$

which defines a set of trajectories (or ‘flow lines’) in the $(\Omega_m, \Omega_\Lambda)$ -plane. This equation also highlights the significance of the points $(1, 0)$ and $(0, 1)$ in this plane, which act as ‘attractors’ for the trajectories. This is illustrated in Figure 15.10, which shows a set of trajectories for various cosmological models. Since any general point in the plane defines a unique trajectory passing through that point, it is convenient to specify each trajectory by the present-day values $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$ (although one could equally well use the values at any other cosmic time). In the left-hand panel, we plot trajectories passing through $\Omega_{m,0} = 0.3$ and $\Omega_{\Lambda,0} = 0.1, 0.2, \dots, 1.1$, and in the right-hand panel the trajectories pass through $\Omega_{\Lambda,0} = 0.7$ and $\Omega_{m,0} = 0.1, 0.2, \dots, 1.1$. We see that the trajectories all start at $(1, 0)$, which is an unstable fixed point, and converge on $(0, 1)$, which is a stable fixed point.

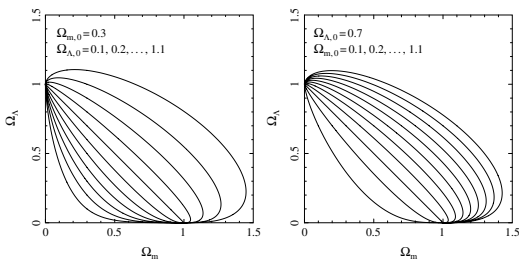


Figure 15.10 Evolution of the density parameters Ω_m and Ω_Λ for various cosmological models passing through the points $\Omega_{m,0} = 0.3$ and $\Omega_{\Lambda,0} = 0.1, 0.2, \dots, 1.1$ (left-hand panel) and $\Omega_{\Lambda,0} = 0.7$ and $\Omega_{m,0} = 0.1, 0.2, \dots, 1.1$ (right-hand panel).

It is worth noting the profound effect of a non-zero cosmological constant on the evolution of the density parameters. In the case $\Lambda = 0$, any slight deviation from $\Omega_m = 1$ in the early universe results in a rapid evolution away from the point $(1, 0)$ along the Ω_m -axis, tending to $(0, 0)$ for an open universe and to $(\infty, 0)$ for a closed one. If $\Lambda > 0$, however, the trajectory is 'refocussed' and tends to the spatially flat de Sitter case $(0, 1)$. Indeed, for a wide range of initial conditions, by the time the matter density has reached $\Omega_m \approx 0.3$ the universe is close to spatially flat.

15.11 Evolution of the spatial curvature

We may investigate directly the behaviour of the spatial curvature as the universe expands by determining the evolution of the curvature density parameter

$$\Omega_k = 1 - \Omega_m - \Omega_r - \Omega_\Lambda = -\frac{c^2 k}{H^2 R^2}. \quad (15.47)$$

Differentiating the final expression on the right-hand side with respect to cosmic time, or combining the derivatives (15.46), one quickly finds that

$$\dot{\Omega}_k = 2\Omega_k H q = \Omega_k H (\Omega_m + 2\Omega_r - 2\Omega_\Lambda), \quad (15.48)$$

where q is the deceleration parameter. We observe that if $\Omega_\Lambda = 0$ then the quantity in parentheses is always positive. Thus, in this case, if Ω_k differs slightly from zero at some early cosmic time then the spatial curvature rapidly evolves away from the spatially flat case. In particular, $\Omega_k \rightarrow 1$ in the open case and $\Omega_k \rightarrow -\infty$ in the closed case. The presence of a positive cosmological constant, however, changes this behaviour completely. In this case, at some finite cosmic time the $2\Omega_\Lambda$ term in (15.48) will dominate the matter and radiation terms, with the result that Ω_k is 'refocussed' back to $\Omega_k = 0$.

We may in fact obtain an analytic expression for the spatial curvature as a function of redshift z , in terms of the present-day values of the density parameters. Substituting for $c^2 k$ from (15.47) evaluated at $t = t_0$, and noting that $R_0/R = 1 + z$, we obtain the useful general formula

$$\Omega_k(z) = \left[\frac{H_0(1+z)}{H(z)} \right]^2 \Omega_{k,0}.$$

Using our expression (15.38) for $H(z)$ then gives

$$\Omega_k(z) = \frac{\Omega_{k,0}}{\Omega_{m,0}(1+z) + \Omega_{r,0}(1+z)^2 + \Omega_{\Lambda,0}(1+z)^{-2} + \Omega_{k,0}}.$$

In particular, we see that (apart from models with only vacuum energy), even if the present-day value $\Omega_{k,0}$ differs greatly from zero, at very high redshift (i.e. in the distant past) $\Omega_k(z)$ must have differed by only a tiny amount from zero. Since today we measure the value $\Omega_{k,0}$ to be (conservatively) in the range -0.5 to 0.5 , this means that at very early epochs Ω_k must have been very finely tuned to near zero. This tuning of the initial conditions of the expansion is called the *flatness problem* and has no solution within standard cosmological models. From our above discussion, however, the presence of a positive cosmological constant goes some way to explaining why the universe is close to spatially flat at the present epoch.

15.12 The particle horizon, event horizon and Hubble distance

Thus far, we have considered the evolution of the entire spatial part of the FRW geometry. It is, however, interesting to consider the extent of the region ‘accessible’ (via light signals) to some comoving observer at a given cosmic time t .

Particle horizon

Let us consider a comoving observer O situated (without loss of generality) at $\chi = 0$. Suppose further that a second comoving observer E has coordinate χ_1 and emits a photon at cosmic time t_1 , which reaches O at time t . Assuming light to be the fastest possible signal, the only signals emitted at time t_1 that O receives by the time t are from radial coordinates $\chi < \chi_1$.

The comoving coordinate χ_1 of the emitter E is determined by

$$\chi_1 = c \int_{t_1}^t \frac{d\bar{t}}{R(\bar{t})}. \quad (15.49)$$

If the integral on the right-hand side diverges as $t_1 \rightarrow 0$ then χ_1 can be made as large as we please by taking t_1 sufficiently small. Thus, in this case, in principle it is possible to receive signals emitted at sufficiently early epochs from *any* comoving particle (such as a typical galaxy). If, however, the integral converges as $t_1 \rightarrow 0$ then χ_1 can never exceed a certain value for a given t . In this case our vision of the universe is limited by a *particle horizon*. At any given cosmic time t , the χ -coordinate of the particle horizon is given by

$$\chi_p(t) = c \int_0^t \frac{d\bar{t}}{R(\bar{t})} = c \int_0^{R(t)} \frac{dR}{R\dot{R}}, \quad (15.50)$$

where in the second equality we have rewritten the expression as an integral over R . The corresponding proper distance to the particle horizon is $d_p(t) = R(t)\chi_p(t)$.

We see that expression (15.50) will be *finite* if $R\dot{R} \sim R^\alpha$ with $\alpha < 1$, which is equivalent to the condition $\ddot{R} < 0$. Hence, *any* universe for which the expansion has been continually *decelerating* up to the cosmic time t will have a finite particle horizon at that time. Clearly, this includes all the Friedmann models that we discussed earlier, but particle horizons also occur in other cosmological models, for example in the spatially flat Lemaitre model with $\Omega_{m,0} \approx 0.3$ and $\Omega_{\Lambda,0} \approx 0.7$ that seems to provide a reasonable description of our universe.

On differentiating (15.50) with respect to t , we have $d\chi_p/dt = c/R(t)$, which is always greater than zero. Thus, the particle horizon of a comoving observer grows as the cosmic time t increases, and so parts of the universe that were not in view previously must gradually come into view. This does not mean, however, that a galaxy that was not visible at one instant suddenly appears in the sky a moment later! To understand this, we note that if the universe has a big-bang origin then we have $R(t_1) \rightarrow 0$ as $t_1 \rightarrow 0$, and so $z \rightarrow \infty$. Thus, the particle horizon at any given cosmic time is the surface of *infinite redshift*, beyond which we cannot see. If the particle horizon grew to encompass a galaxy, the galaxy would therefore appear at first with an infinite redshift, which would gradually reduce as more cosmic time passed. Hence the galaxy would not simply ‘pop’ into view.⁷

In fact, we can obtain explicit expressions for the particle horizon in some cosmological models. For example, a matter-dominated model at early epochs obeys $R(t)/R_0 = (t/t_0)^{2/3}$, whereas a radiation-dominated model at early epochs obeys $R(t)/R_0 = (t/t_0)^{1/2}$. Substituting these expressions into (15.50) gives the proper distance to the particle horizon at cosmic time t as

$$d_p(t) = 3ct \quad (\text{matter-dominated}), \quad d_p(t) = 2ct \quad (\text{radiation-dominated}).$$

These proper distances are larger than ct because the universe has expanded while the photon has been travelling. Alternatively, if one has an analytic expression for $\chi(z)$ for some cosmological model then the corresponding expression for χ_p may be obtained simply by letting $z \rightarrow \infty$.

The existence of particle horizons for the common cosmological models illustrates the *horizon problem*, i.e. how do vastly separated regions display the same physical characteristics (e.g. the nearly uniform temperature of the cosmic microwave background) when, according to standard cosmological models, these regions could never have been in causal contact? This problem, like the flatness problem, is a serious challenge to standard cosmology that can only be resolved by invoking the theory of inflation (see Chapter 16).

⁷ In practice, our view of the universe is not limited by our particle horizon but by the epoch of recombination, which occurred at $z_{\text{rec}} \approx 1500$ (long before the formation of any galaxies). Prior to this epoch, the universe was ionised and photons were frequently scattered by the free electrons, whereas after this point electrons and protons (and neutrons) combined to form atoms and the photons were able to propagate freely. This *surface of last scattering* is therefore the effective limit of our observable universe.

The horizon problem can be illustrated by a simple example. Consider a galaxy at a proper distance of 10^9 light years away from us. Since the age of the universe is $\sim 1.5 \times 10^{10}$ years, there has been sufficient time to exchange about 15 light signals with the galaxy. At earlier times, when the scale factor R was smaller, everything was closer together and so we might have naively expected that this would improve causal contact. In a continuously decelerating universe, however, it makes the problem *worse*. At, for example, the epoch of *recombination* (when the cosmic microwave background photons were emitted) the redshift z was approximately 1000, so $R(t_{\text{rec}})/R_0 \approx 10^{-3}$ and the proper distance to the ‘galaxy’ is 10^6 light years.⁸ If we assume, for simplicity, that after t_{rec} the expansion followed a matter-dominated Einstein–de-Sitter universe, then

$$\left(\frac{t_{\text{rec}}}{t_0}\right)^{2/3} = \frac{R_{\text{rec}}}{R_0} = 10^{-3},$$

and so $t_{\text{rec}} = 1.5 \times 10^{5.5}$ years. However, assuming that prior to t_{rec} the expansion followed a radiation-dominated Einstein–de-Sitter model, the proper distance to the particle (causal) horizon is $2ct_{\text{rec}} = 3 \times 10^{5.5}$ light years. Thus, by t_{rec} ‘we’ could not have exchanged even *one* light signal with the other ‘galaxy’.

Event horizon

Although our particle horizon grows as the cosmic time t increases, in some cosmological models there could be events that we may *never* see (or, conversely, never influence). Returning to our expression (15.49), we see that if the integral on the right-hand side diverges as $t \rightarrow \infty$ (or the time at which R equals zero again), then it will be possible to receive light signals from any event. However, if the integral instead converges for large t then, for light signals emitted at t_1 , we will only ever receive those from events for which the χ -coordinate is less than

$$\chi_e(t_1) = c \int_{t_1}^{t_{\text{max}}} \frac{dt}{R(t)},$$

where t_{max} is either infinity or the time of the big crunch (i.e. $R(t_{\text{max}}) = 0$). This is called the *event horizon*. By symmetry, $\chi_e(t_0)$ is the maximum χ -coordinate that can be reached by a light signal sent by us today.

Hubble distance

From our discussion in Section 15.7, the elapsed cosmic time t since the big bang is, in general, of the order $H^{-1}(t)$, which is known as the Hubble time and

⁸ In reality the galaxy would not yet have formed, but this does not affect the main point of the argument.

provides a characteristic time scale for the expansion of the universe. In a similar way, at a cosmic time t one can define the *Hubble distance*

$$d_H(t) = cH^{-1}(t),$$

which provides a characteristic length scale for the universe. We may also define the *comoving Hubble distance*

$$\chi_H(t) = \frac{d_H(t)}{R(t)} = \frac{c}{H(t)R(t)} = \frac{c}{\dot{R}(t)}, \quad (15.51)$$

where in the last equality we have used the fact that $H = \dot{R}/R$. The above expression simply gives the χ -coordinate corresponding to the Hubble distance.

The Hubble distance $d_H(t)$ corresponds to the typical length scale (at cosmic time t) over which physical processes in the universe operate coherently. It is also the length scale at which general-relativistic effects become important; indeed, on length scales much less than $d_H(t)$, Newtonian theory is often sufficient to describe the effects of gravitation. From our discussion above, we further note that the proper distance to the particle horizon for standard cosmological models is typically

$$d_p(t) \sim ct \sim cH^{-1}(t).$$

Thus, we see that the particle horizon in such cases is of the same order as the Hubble distance. As a result, the Hubble distance is often described simply as the ‘horizon’. It should be noted, however, that the particle horizon and the Hubble distance are distinct quantities, which may differ by many orders of magnitude in *inflationary cosmologies*, which we discuss in the next chapter. In particular, we note that the particle horizon at time t depends on the entire expansion history of the universe to that point, whereas the Hubble distance is defined instantaneously at t . Moreover, once an object lies within an observer’s particle horizon it remains so. On the contrary, an object can be within an observer’s Hubble distance at one time, lie outside it at some later time and even come back within it at a still later epoch.

Exercises

- 15.1 For blackbody radiation, the number density of photons with frequencies in the range $[\nu, \nu + d\nu]$ is given by

$$n(\nu, T) d\nu = \frac{8\pi\nu^2}{c^3 (e^{h\nu/kT} - 1)} d\nu, \quad (\text{E15.1})$$

where T is the ‘temperature’ of the radiation. By conserving the total number of photons, show that the photon energy distribution of the cosmic microwave

background (CMB) radiation retains its general blackbody form as the universe expands. Show further that the total number density n of photons is

$$n(T) = 0.244 \left(\frac{2\pi k_B T}{hc} \right)^3.$$

Hence show that the present-day number density of CMB photons in the universe is $n_0 \approx 4 \times 10^8 \text{ m}^{-3}$, and compare this with the present-day number density of protons. How does this ratio vary with cosmic time?

Hint: $\int_0^\infty \frac{x^2}{e^x - 1} dx = 0.244\pi^2$.

- 15.2 Suppose that the present-day energy densities of radiation and matter (in the form of dust) are $\rho_r(t_0)c^2$ and $\rho_m(t_0)c^2$ respectively. Show that the energy densities of the two components were equal at a redshift z_{eq} given by

$$1 + z_{\text{eq}} = \frac{\rho_m(t_0)}{\rho_r(t_0)}.$$

What assumptions underlie this result? Hence show that

$$1 + z_{\text{eq}} = \frac{3c^2 \Omega_{m,0} H_0^2}{8\pi G a T_0^4},$$

where a is the reduced Stefan–Boltzmann constant and T_0 is the present-day temperature of the cosmic microwave background. Show that for our universe $z_{\text{eq}} \approx 5000$. What was the temperature of the CMB radiation at this epoch?

- 15.3 Show that in the early, radiation-dominated, phase of the universe, the temperature T of the radiation satisfies the equation

$$\left(\frac{\dot{T}}{T} \right)^2 = \frac{8\pi G a T^4}{3},$$

where the dot denotes differentiation with respect to the cosmic time t and a is the reduced Stefan–Boltzmann constant. Hence show that

$$T = \left(\frac{3c^2}{32\pi G a} \right)^{1/4} t^{-1/2} \approx 1.5 \times 10^{10} \left(\frac{t}{s} \right)^{-1/2} \text{ K},$$

and that the cosmic time at matter–radiation equality is $t_{\text{eq}} \approx 16\,000$ years.

- 15.4 The CMB radiation was emitted at the epoch of recombination at redshift $z_{\text{rec}} \approx 1500$. Show that $t_{\text{rec}} \approx 450\,000$ years.
- 15.5 Consider a cylindrical piston chamber of cross-sectional area A ‘filled’ with vacuum energy. The piston is withdrawn a linear distance dx . Show that the energy created by withdrawing the piston equals the work done by the vacuum, provided that

$$p_{\text{vac}} = -\rho_{\text{vac}}c^2.$$

Hence show that, in this case, the vacuum energy density is constant as the piston is withdrawn.

- 15.6 Show that the present-day value of the scale factor of the universe may be written as

$$R_0 = \frac{c}{H_0} \left(\frac{k}{\Omega_{k,0}} \right)^{1/2}.$$

What value does R_0 take in a spatially flat universe?

- 15.7 Show that, for our universe to be spatially flat, the total density must be equivalent to ≈ 5 protons m^{-3} .
- 15.8 In the Newtonian cosmological model discussed in Exercise 14.14, show that the total energy E of the test particle of mass m can be written as

$$E = \frac{1}{2}m(1 - \Omega_m)R^2H^2,$$

and interpret this result physically.

- 15.9 Show that at all cosmic times the density parameters obey the relation

$$\Omega_m + \Omega_r + \Omega_\Lambda + \Omega_k = 1.$$

- 15.10 In terms of the dimensionless density parameters, show that the two cosmological field equations can be written in the forms

$$H^2 = H_0^2 [\Omega_{m,0} a^{-3} + \Omega_{r,0} a^{-4} + \Omega_{\Lambda,0} + \Omega_{k,0} a^{-2}],$$

$$q = \frac{1}{2}(\Omega_m + 2\Omega_r - 2\Omega_\Lambda),$$

where H and q are the Hubble and deceleration parameters respectively, and $a = R/R_0$ is the normalised scale factor.

- 15.11 The *conformal time* variable is defined by $d\eta = c dt/R$. Hence show that the second cosmological field equation can be written as

$$\left(\frac{da}{d\eta} \right)^2 = -\frac{k}{\Omega_{k,0}} (\Omega_{m,0} a + \Omega_{r,0} + \Omega_{\Lambda,0} a^4 + \Omega_{k,0} a^2).$$

- 15.12 Show that the density parameter for matter, radiation or the vacuum varies with the normalised scale factor as

$$\Omega_i = \Omega_{i,0} \left(\frac{H_0}{H} \right)^2 a^{-3(1+w_i)},$$

where w_i is the appropriate equation-of-state parameter.

- 15.13 Show that the condition for the $a(t)$ -curve to have a turning point is

$$f(a) \equiv \Omega_{\Lambda,0} a^3 + (1 - \Omega_{m,0} - \Omega_{\Lambda,0})a + \Omega_{m,0} = 0.$$

In the case $\Omega_{\Lambda,0} > 0$, show by evaluating the derivatives $f'(a)$ and $f''(a)$ that the condition for $f(a)$ to have a single positive root at $a = a_*$ is $f(a_*) = f'(a_*) = 0$. Show further that this root occurs at

$$a_* = \left(\frac{\Omega_{m,0}}{2\Omega_{\Lambda,0}} \right)^{1/3}.$$

Hence show that the values $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$, along any dividing line in this plane that separates those models with a turning point in the $a(t)$ -curve from those without, must satisfy

$$4(1 - \Omega_{m,0} - \Omega_{\Lambda,0})^3 + 27\Omega_{m,0}^2\Omega_{\Lambda,0} = 0.$$

- 15.14 Show that the substitution $x = [\Omega_{\Lambda,0}/(4\Omega_{m,0})]^{1/3}$ reduces the final cubic equation in Exercise 15.13 to

$$x^3 - \frac{3x}{4} + \frac{\Omega_{m,0} - 1}{4\Omega_{m,0}} = 0.$$

By using the standard formulae for the roots of a cubic, or otherwise, verify the results (15.18–15.20).

- 15.15 Show that, in terms of the variable $\hat{t} = H_0(t - t_0)$, the evolution of the normalised scale factor obeys the equation

$$\left(\frac{da}{d\hat{t}}\right)^2 = \Omega_{m,0}a^{-1} + \Omega_{r,0}a^{-2} + \Omega_{\Lambda,0}a^2 + 1 - \Omega_{m,0} - \Omega_{r,0} - \Omega_{\Lambda,0}.$$

Show that, when one is integrating this equation numerically, an iterative algorithm of the form

$$a_{n+1} \approx a_n + \left(\frac{da}{d\hat{t}}\right)_n \Delta\hat{t}$$

would not be able to propagate the solution through points for which $da/d\hat{t} = 0$.

- 15.16 For a $k = -1$ Friedmann model containing no matter or radiation, show that the line element becomes

$$ds^2 = c^2 dt^2 - c^2 t^2 [d\chi^2 + \sinh^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2)].$$

Show that this metric describes a Minkowski spacetime.

- 15.17 For a dust-only Friedmann model with $\Omega_{m,0} > 1$, show that

$$a = \frac{\Omega_{m,0}}{2(\Omega_{m,0} - 1)}(1 - \cos \psi), \quad t = \frac{\Omega_{m,0}}{2H_0(\Omega_{m,0} - 1)^{3/2}}(\psi - \sin \psi).$$

Hence show that the $a(t)$ -curve has a maximum at

$$a_{\max} = \frac{\Omega_{m,0}}{\Omega_{m,0} - 1}, \quad t_{\max} = \frac{\pi}{2H_0} \frac{\Omega_{m,0}}{(\Omega_{m,0} - 1)^{3/2}},$$

and that the age t_0 of such a universe is given by

$$t_0 = \frac{\Omega_{m,0}}{2H_0(\Omega_{m,0} - 1)^{3/2}} \left[\cos^{-1} \left(\frac{2}{\Omega_{m,0}} - 1 \right) - \frac{2}{\Omega_{m,0}} (\Omega_{m,0} - 1)^{1/2} \right] < \frac{2}{3H_0}.$$

- 15.18 For the Einstein–de-Sitter model, prove the following useful results:

$$a(t) = \left(\frac{t}{t_0}\right)^{2/3}, \quad H(t) = \frac{2}{3t} = H_0(1+z)^{3/2}, \quad q_0 = \frac{1}{2}, \quad \rho_m(t) = \frac{1}{6\pi G t^2}.$$

15.19 For a radiation-only Friedmann model with $\Omega_{r,0} \neq 1$, show that

$$a(t) = (2H_0\Omega_{r,0}^{1/2}t)^{1/2} \left(1 + \frac{1 - \Omega_{r,0}}{2\Omega_{r,0}^{1/2}}H_0t\right)^{1/2}.$$

Hence, for $\Omega_{r,0} > 1$, show that the $a(t)$ -curve has a maximum at

$$a_{\max} = \left(\frac{\Omega_{r,0}}{\Omega_{r,0} - 1}\right)^{1/2}, \quad t_{\max} = \frac{1}{H_0} \frac{\Omega_{r,0}^{1/2}}{(\Omega_{r,0} - 1)},$$

and that the age t_0 of such a universe is given by

$$t_0 = \frac{1}{H_0} \frac{1}{\Omega_{r,0}^{1/2} + 1} < \frac{1}{2H_0}.$$

15.20 For the spatially flat, radiation-only, Friedmann model, prove the following useful results:

$$a(t) = \left(\frac{t}{t_0}\right)^{1/2}, \quad H(t) = \frac{1}{2t} = H_0(1+z)^2, \quad q_0 = 1, \quad \rho_r(t) = \frac{3}{32\pi G t^2}.$$

15.21 For a spatially flat Friedmann model containing both matter and radiation, show that

$$H_0 t = \frac{2}{3\Omega_{m,0}^2} \left[(\Omega_{m,0}a + \Omega_{r,0})^{1/2} (\Omega_{m,0}a - 2\Omega_{r,0}) + 2\Omega_{r,0}^{3/2} \right].$$

15.22 For a Lemaitre model containing no radiation, show that at the point of inflection of the $a(t)$ -curve the value of the normalised scale factor is

$$a_* = \left(\frac{\Omega_{m,0}}{2\Omega_{\Lambda,0}}\right)^{1/3},$$

and calculate a_* for our universe. Show further that, in the vicinity of the point of inflection, the scale factor obeys the equation

$$\dot{a}^2 \approx H_0^2 [\Omega_{k,0} + 3\Omega_{\Lambda,0}a_*^2 + 3\Omega_{\Lambda,0}(a - a_*)^2]$$

and that this has the solution

$$a(t) = a_* + a_* \left[1 + \frac{1}{3}\Omega_{k,0} \left(\frac{1}{4}\Omega_{\Lambda,0}\Omega_{m,0}^2\right)^{-1/3} \right]^{1/2} \sinh [H_0(3\Omega_{\Lambda,0})^{1/2}(t - t_*)].$$

15.23 For a spatially flat Lemaitre model containing no radiation, show that

$$H_0 t = \frac{2}{3\sqrt{|\Omega_{\Lambda,0}|}} \int_0^{\sqrt{a^3|\Omega_{\Lambda,0}|/(1-\Omega_{\Lambda,0})}} \frac{dy}{\sqrt{1 \pm y^2}}.$$

Hence show that

$$a(t) = \left(\frac{1 - \Omega_{\Lambda,0}}{|\Omega_{\Lambda,0}|}\right)^{1/3} \begin{cases} \sinh^{2/3} \left(\frac{3}{2}\sqrt{\Omega_{\Lambda,0}}H_0t\right) & \text{if } \Omega_{\Lambda,0} > 0, \\ \sin^{2/3} \left(\frac{3}{2}\sqrt{|\Omega_{\Lambda,0}|}H_0t\right) & \text{if } \Omega_{\Lambda,0} < 0. \end{cases}$$

15.24 Show that, in general,

$$\frac{\ddot{R}}{R} = H^2 + \dot{H}.$$

Hence use the cosmological field equations to show that, for a spatially flat Lemaitre model containing no radiation, the Hubble parameter and the matter density satisfy the equations

$$2\dot{H} + 3H^2 = \Lambda c^2,$$

$$3H^2 - \Lambda c^2 = 8\pi G\rho_m.$$

Assuming $\Lambda > 0$ and requiring $\rho_m > 0$, thus show that

$$H(t) = \sqrt{\frac{\Lambda c^2}{3}} \coth\left(\frac{3}{2}\sqrt{\frac{\Lambda c^2}{3}}t\right),$$

$$\rho_m(t) = \frac{\Lambda c^2}{8\pi G} \operatorname{cosech}^2\left(\frac{3}{2}\sqrt{\frac{\Lambda c^2}{3}}t\right),$$

and therefore find expressions for $\Omega_m(t)$ and $\Omega_\Lambda(t)$. Show further that

$$t = \frac{2}{3H} \frac{\tanh^{-1}\sqrt{\Omega_\Lambda}}{\sqrt{\Omega_\Lambda}}.$$

Hint: $\int a^2/(a^2 - x^2) dx = \coth^{-1}(x/a) + \text{constant}$, for $x^2 > a^2$.

15.25 Show that for a physically reasonable perfect fluid (i.e. density > 0 and pressure ≥ 0) there is no static isotropic homogeneous solution to Einstein's equations with $\Lambda = 0$. Show that it is possible to obtain a static zero-pressure solution by the introduction of a cosmological constant Λ such that

$$\Lambda c^2 = 4\pi G\rho_{m,0} = \frac{c^2 k}{R_0^2}.$$

Show that this solution is unstable, however.

15.26 Show that the comoving χ -coordinate of a galaxy emitting a photon at time t that is received at t_0 is given by

$$\chi = \frac{c}{R_0} \int_{(1+z)^{-1}}^1 \frac{da}{a\dot{a}}.$$

Using the cosmological field equation (15.13) to substitute for $\dot{a}$, show that

$$\chi(z) = \frac{c}{R_0 H_0} \int_{(1+z)^{-1}}^1 \frac{dx}{\sqrt{\Omega_{m,0}x + \Omega_{r,0} + \Omega_{\Lambda,0}x^4 + \Omega_{k,0}x^2}}.$$

15.27 For a dust-only Friedmann model, show that the luminosity-distance relation varies with redshift as

$$d_L(z) = \frac{2c}{H_0 \Omega_{m,0}^2} \left[\Omega_{m,0} z + (\Omega_{m,0} - 2) \left(\sqrt{\Omega_{m,0} z + 1} - 1 \right) \right].$$

This result is known as *Mattig's formula*.

15.28 For a Friedmann model dominated by a single source of energy density, show that

$$\Omega_i^{-1}(z) - 1 = \frac{\Omega_{i,0}^{-1} - 1}{(1+z)^{1+3w_i}},$$

where w_i is the equation-of-state parameter of the source. Use this result to comment on the flatness problem.

15.29 For a general cosmological model, show that

$$\dot{\Omega}_i = \Omega_i H (\Omega_m + 2\Omega_r - 2\Omega_\Lambda - 1 - 3w_i),$$

where i denotes matter, radiation or the vacuum.

15.30 By differentiating the definition $\Omega_k = -kc^2/(R^2H^2)$, show that

$$\dot{\Omega}_k = 2\Omega_k Hq = \Omega_k H (\Omega_m + 2\Omega_r - 2\Omega_\Lambda),$$

where q is the deceleration parameter.

15.31 Show that the particle horizon at cosmic time t is given

$$\chi_p(t) = \frac{c}{R_0 H_0} \int_0^{a(t)} \frac{dx}{\sqrt{\Omega_{m,0}x + \Omega_{r,0} + \Omega_{\Lambda,0}x^4 + \Omega_{k,0}x^2}}.$$

15.32 Consider the cosmological line element

$$ds^2 = c^2 dt^2 - e^{2t/b} (dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2).$$

Light signals from a galaxy at coordinate distance r are emitted at epoch t_1 and received by an observer at epoch t_0 . Show that

$$\frac{r}{bc} = e^{-t_1/b} - e^{-t_0/b}.$$

For a given r , show that there is a maximum epoch t_1 and interpret this result physically. Show that a light ray emitted by the observer asymptotically approaches the coordinate $r = bc$ but never reaches it.